MOTIDONE TABLET 10MG

DESCRIPTION: An oblong shaped tablet, pink in colour with marking "DUO".

COMPOSITION: Each tablet contains Domperidone maleate equivalent to Domperidone 10 mg.

PHARMACODYNAMICS: Pharmacologically, domperidone is a dopamine antagonist with antiemetic properties similar to those of metoclopramide and certain neuroleptic drugs. Unlike these drugs, however, domperidone does not readily cross the blood brain barrier. It seldom causes extra-pyrimidal side effects, but does cause a rise in prolactin levels. Its antiemetic effect may be due to a combination of peripheral (gastro-kinetic) effects and antagonism of central dopamine receptors in the chemo-receptor trigger zone which lies in the area postrema and is regarded as being outside the blood brain barrier. Animal studies have shown that domperidone has no effect on plasma concentrations of homovanillic acid, a metabolite of dopamine. It also antagonises the behavioural effects of dopamine much more effectively when administered intracerebrally than when given systemically. These findings, together with the low concentrations found in the brain, indicate a predominantly peripheral effect of domperidone on dopamine receptors.

PHARMACOKINETICS:

Domperidone is rapidly absorbed following intramuscular and oral administration with peak plasma concentrations occurring at approximately 10 and 30 minutes respectively. Systemic bioavailability of oral domperidone is 13 to 17%. The low oral bioavailability is probably due to "first-pass" gut wall and hepatic metabolism. Oral bioavailability is decreased by prior administration of cimetidine or sodium bicarbonate. The time of peak absorption is slightly delayed and the AUC somewhat increased when the oral drug is taken after a meal.

Oral domperidone does not appear to accumulate or induce its own metabolism; a peak plasma level after 90 minutes of 21 ng/ml after two weeks oral administration of 30 mg per day was almost the same as that of 18 ng/ml after the first dose. Domperidone is 91-93% bound to plasma proteins.

Distribution studies with radiolabelled drug in animals have shown wide tissue distribution with low brain concentration. Small amounts of drug cross the placenta in rats and the drug is excreted in the breast milk of this species.

Domperidone undergoes rapid and extensive hepatic metabolism by hydroxylation and N-dealkylation. Urinary and faecal excretion amounts to 31 and 66% respectively of the oral dose. The proportion of the drug excreted unchanged is small (10% of faecal excretion and approximately 1% of urinary excretion).

The plasma half-life after a single oral dose is 7-9 hours in healthy subjects but is prolonged in patients with severe renal insufficient

INDICATIONS:

The dyspeptic symptom complex that is often associated with delayed gastric emptying, gastro-oesophageal reflux and oesophagitis:

- epigastric sense of fullness, early satiety, feeling of abdominal distension, upper abdominal pain;
- bloating, eructation, flatulence;
- nausea and vomiting;
- heartburn with or without regurgitations of gastric contents in the mouth.

RECOMMENDED DOSAGE:

Domperidone should be taken 15-30 minutes before meals, and if necessary, before retiring.

Adults: 10 mg three to four times daily. If necessary, this dose can be doubled after two weeks if insufficient response.

In patients with renal insufficiency: (creatinine serum levels > 6 mg/100 mL ie. > 0.6 mmol/L) the elimination half-life drug of domperidone was increased from 7.4 to 20.8 hours but plasma drugs were lower than in healthy volunteers. Since very little unchanged drug is excreted via the kidneys, it is unlikely that the dose needs to be adjusted for single acute administration in patients with renal insufficiency. However, on repeated administration, the dosing frequency will need to be reduced to once or twice daily depending on the severity of the impairment, and the dose may need to be reduced.

Food: Oral domperidone is recommended to be taken before meals. If taken after meals, absorption of the drug is somewhat delayed.

ROUTE OF ADMINISTARTION: Oral

CONTAINDICATIONS: Domperidone should not be used whenever stimulation of gastric motility might be dangerous e.g. in the presence of gastrointestinal haemorrhage, mechanical obstruction or perforation. Domperidone is also contraindicated in the following:

- in patients with known intolerance to the drug
- in patients with a prolactin-releasing pituitary tumour (prolactinoma)
- prophylaxis of post-operative vomiting or for chronic administration

WARNING & PRECAUTIONS:

Prolactin levels: There are limited safety data in long term use (ie. beyond six months) of domperidone. However, it has been shown to produce an increase in plasma prolactin. The raised level persists with chronic administration but falls to normal on discontinuing the drug.

It has been shown in vitro that about 1/3 of breast cancers are prolactin dependent and hence this point must be considered when prescribing for a patient with a past history of breast cancer.

Endocrine disturbances such as galactorrhoea, amenorrhoea, gynaecomastia and impotence have been reported with drugs which stimulate prolactin release.

Use with impaired hepatic function: Since domperidone is extensively metabolised in the liver, it should be used with caution in patients with hepatic impairment.

INTERACTIONS WITH OTHER MEDICAMENTS:

Concomitant administration of anticholinergic drugs may antagonise the anti-dyspeptic affect of domperidone. If administered prior to atropine, domperidone reduces the relaxant effect of atropine upon the lower oesophageal sphincter, but has no reversing effect if atropine is administered first. Since domperidone has gastro-kinetic effects it could influence the absorption of concomitantly orally administered drugs, particularly those of sustained release or enteric coated formulations. However, in patients already stabilized on digoxin, paracetamol or haloperidol concomitant administration of domperidone did not influence the blood levels of these drugs.

Concomitant use with antacids and antisecretory drugs with domperidone will reduce its oral bioavailability. Dosing with these 2 agents should be separated from dosing of domperidone by at least 2 hours.

In patients stabilised on digoxin, paracetamol or haloperidol administration of domperidone did not influence the blood levels of these drugs.

Domperidone has been used with neuroleptics without potentiation of their activity.

Dopaminergic: Antagonism of hypoprolactin effect of bromocriptine.

USE IN PRENANCY & LACTATION:

Use in pregnancy: There are no adequate and well controlled studies in pregnant women and hence this drug should be used during pregnancy only if clearly needed.

Use in lactation: The drug is excreted in breast milk of lactating rats and probably also in humans. It is therefore not recommended for nursing mothers unless the expected benefits outweigh any potential risk.

SIDE EFFECTS:

Mild and transient abdominal cramps and dry mouth have been reported rarely. There have been no extrapyramidal side effects reported during controlled therapeutic trials although a few anecdotal reports of such effects have been published.

- May induce an increase in the plasma prolactin level which may be associated with galactorrhoea and less frequently with gynaecomastia.
- Rashes and other allergic reactions.
- Acute dystonic reactions.

Central Nervous System: Dry mouth, headache, insomnia, dizziness, thirst, lethargy, irritability, extrapyramidal reactions (rare).
Gastrointestinal: Abdominal cramps, diarrhoea, regurgitation, changes in appetite, nausea, heartburn, constipation.
Dermatologic: Rash, pruritus, urticaria.
Urinary: Urinary frequency, dysuria.
Cardiovascular: Oedema, palpitations.
Musculoskeletal: Leg cramps, asthenia.
Other: Conjunctivitis, stomatitis, drug intolerance.

There have been reports of adverse effects possibly related to increases in serum prolactin. These effects include: Gynecomastia, breast tenderness, swelling of the breasts, irregular menses, amenorrhoea, a decrease or loss of libido, breast secretion and lactation. These effects occurred in patients who received up to 120 mg per day in four divided doses.

SYMPTOMS AND TREATMENT OF OVERDOSE:
Overdosage: Symptoms of overdosage may, include drowsiness, disorientation and extrapyramidal reactions, particularly in children. Acute toxicity studies in animals resulted in primarily central nervous system symptoms.

Treatment:
- Gastric lavage
- Activated charcoal and close observation of the patient are recommended.
- anticholinergic, anti-Parkinson drugs or antihistamines with anticholinergic properties may be helpful in controlling the extrapyramidal reactions.

PACK SIZE: Blister pack of 50 x 10’s and 10 x 10’s tablets in a box.

STORAGE CONDITIONS: Store below 25°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
DUOPHARMA (M) SDN BHD
Lot 2599 Jalan Seruling 59 Kawasan 3
Taman Klang Jaya, 41200 Klang
Selangor Darul Ehsan, MALAYSIA